



THE HARNESS ENGINEERING & MODEL WRANGLING HACKATHON

RESOURCE GUIDE

HACKATHON DAY — SEPT 26 · The Malin Chelsea

MONGODB.LOCAL NYC — SEPT 30 · Pier 36

PROBLEM STATEMENTS

This hackathon focuses on harness engineering: building the systems around a model (rules, context policies, guardrails, tool access and memory) that let agents adapt to a task, sustain work over long horizons and act on real data, including databases, documents, code and other business systems. Build against one of the two statements below, with MongoDB as your data and memory layer.

STATEMENT ONE: RECURSIVE HARNESSING

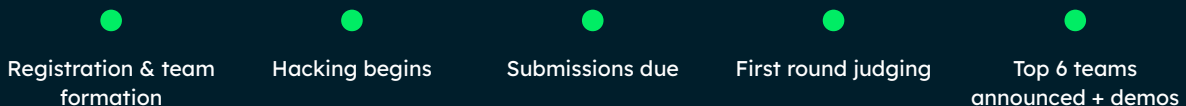
Build a self-improving agent harness that automatically evolves its own architecture: updating rules, context policies, guardrails, tool access, and more. How can agents dynamically adapt their entire environment to fit a specific user, task, or use case?

STATEMENT TWO: LONG HORIZON ENGINEERING

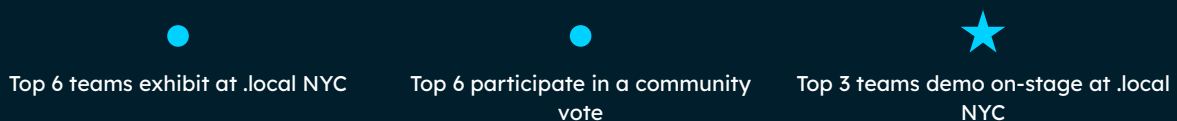
The highest-value agentic work spans hours, days, or weeks. Build a harness that sustains coherent memory across billions of tokens in one session, relentlessly optimizes toward long-term goals, and learns from hard metric signals to complete traditionally difficult tasks.

BUILD IN A DAY. SHOWCASE ON A GLOBAL STAGE.

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HACKATHON REQUIREMENTS

- **Finalist requirement:** You must build your project using the MongoDB Atlas Hackathon Sandbox (sent to email).
- **No prior projects:** All work must be original.
- **Demo access:** Check that your repository is public and your demo link is accessible.
- **Team size:** Up to 4 members.
- **Demo & attendance:** Record an on-site demo Sept 26; at least one team member from finalist teams must be present at MongoDB.local NYC between 10AM – 4:30PM on Sept 30 to be eligible to win.

MONGODB ATLAS SANDBOX

- You should have received an email with a link to join the MongoDB Atlas Sandbox for the hackathon.
- You must create your project + cluster through this link. Keep an eye out for this email!
- **In order to be eligible to be a finalist, your hackathon project must be built in this sandbox.**

CEREBRAL VALLEY PLATFORM

- **All projects must be submitted through the Cerebral Valley Platform.**
- **You will submit:** a link to your public GitHub repository, a 1-minute demo video showing the code & functionality your team built today, and a concise project description detailing what you built.
- **Reminders:** verify that your video has video & audio playback, and double check that all team members have been added to your submission.

PRIZES

1st	2nd	3rd
MongoDB: \$7,000 USD	MongoDB: \$5,000 USD	MongoDB: \$3,000 USD
ElevenLabs: 3 months Scale Tier	ElevenLabs: 3 months Pro Tier	ElevenLabs: 3 months Pro Tier
LangSmith: \$3,000 of Credits	LangSmith: \$2,000 of Credits	LangSmith: \$1,000 of Credits
Vercel: \$1,800 of v0 Credits	Vercel: \$1,200 of v0 Credits	Vercel: \$600 of v0 Credits

MONGODB RESOURCES

RECOMMENDED STARTING POINT

Configure the following before you start building. Together, they give your AI coding assistant the context, connectivity, and prompting guidance it needs to work effectively with MongoDB. Build in the Atlas Hackathon Sandbox from your email; it's required to be a finalist.

- [MongoDB Agent Skills](#) — A set of instructions that can be installed into an AI coding assistant (such as Claude, Cursor, or GitHub Copilot) to provide it with MongoDB best practices for connecting to a database, structuring data, writing queries, and implementing search.
- [MongoDB MCP Server](#) — Enables an AI coding assistant to connect directly to a live MongoDB database to inspect data, run queries, retrieve counts, and manage Atlas resources.
- [Natural Language to MongoDB Queries](#) — Guidance on how to prompt an AI assistant effectively, including what details to provide about the desired task, data structure, and output format.

Recommended sequence: install Agent Skills, connect the MCP Server, and reference the prompting guide throughout development.

DEVELOPER TOOLS

- [Atlas Managed MCP Server](#) — A hosted MCP server that connects AI agents and automated workflows to Atlas using Atlas service accounts instead of end-user credentials. No local MCP infrastructure to run, and you can connect from any harness.
- [Frontier Marketplace Connectors](#) — Native Atlas access inside ChatGPT and Codex, Claude, Gemini, Grok Build, Devin, and Cursor. Query data, inspect schemas and manage indexes without leaving your AI tool.

DATA AND SEARCH

- [Sample Movie Dataset](#) — A ready-to-use dataset that can be loaded in minutes and is preconfigured for AI-powered search.
- [Data Modeling in MongoDB](#) — Best practices for structuring data for efficient use by applications and AI agents.
- [Vector Search](#) — Enables search based on semantic meaning rather than exact keyword matches, forming the basis for long-term, retrievable agent memory.
- [Atlas Search](#) — Provides fast, typo-tolerant keyword search capabilities.
- [Automated Embeddings](#) — Generates and maintains vector embeddings within the database, removing the need for a separate embedding pipeline.
- [Embedding and Reranking API](#) — A single endpoint for embedding and reranking operations, accessible from any application or technology stack.

MEMORY AND AGENTS

- [Building an Agent with Memory and Function Calling](#) — An example of an agent that retains context and executes functions to act on data.

- [Adding Memory to a Chat Application \(Python\)](#) — Demonstrates how to maintain context across a conversation using LangChain and MongoDB Atlas.
- [Adding Memory to a Chat Application \(JavaScript\)](#) — The same approach implemented in JavaScript.

STATE

- [Build an AI Agent with LangGraph and MongoDB Atlas](#) — A practical example using thread-specific checkpoints and conversation persistence.
- [State & Persistence: The Problem of Agent Reliability](#) — A deeper look at checkpoints, suspend/resume, crash recovery, and where state ends and memory begins.

CONTEXT

- [Build AI Agents with MongoDB](#) — Agents that use MongoDB for retrieval alongside APIs and external tools.
- [GraphRAG with MongoDB and LangChain](#) — Ingest external documents and retrieve relationship-aware context.

STARTER CODE

- [GenAI Showcase](#) — A comprehensive library of example applications spanning multiple AI frameworks and programming languages.
- [MongoDB and Python Quickstart](#) — An introductory guide for building with MongoDB Atlas and Python.
- [MERN Starter Application](#) — A starter project using MongoDB, Express, React, and Node.js.

AFTER THE HACKATHON

- [MongoDB for Startups](#) — Get Atlas credits, Voyage AI tokens, technical expertise, and dedicated support on the world's leading AI-ready data platform.

Not sure where to begin? Complete the three items under Recommended starting point, then consult the GenAI Showcase for relevant examples.

PARTNER RESOURCES

Credits and tools from our partners. Most credits are only available to checked-in participants, so make sure you're registered on the Cerebral Valley Platform and checked in on the day.

OPENROUTER

One API key for 500+ AI models, with free credits for the hackathon.

- **Access:** A credit code will be sent to all checked-in participants on the day from 10:30am on Saturday.

ELEVENLABS

AI voice platform covering speech, agents, dubbing and sound effects. One month of the Creator plan free.

- **Access:** Redeem your coupon through the bot in the event Discord once you've checked in. We'll be using the registered & checked-in emails on the Cerebral Valley platform to facilitate this from 10:30am on Saturday.

[ElevenLabs Hacker Guide](#)

LANGCHAIN

\$50 in LangSmith credits plus Deployments access, along with docs and starter templates for building agents.

- **Access:** Redeem through the form below within 10 days of the hackathon, entering the hackathon name as the event. You'll need to add a card to LangSmith to display the credits; it won't be charged.

[Redeem credits](#) · [LangChain hacker resources](#) · [LangGraph docs](#) · [Deep Agents quickstart](#)

AWS — KIRO

AI coding suite (IDE, CLI and Crew) that turns prompts into working, deployed code through spec-driven development.

- **Access:** Free to start with no card or AWS account needed: sign in with Google, GitHub or AWS Builder ID. 50 credits a month on the free tier, plus a 500-credit new-user bonus usable within 30 days.

[Kiro IDE](#) · [Kiro CLI](#) · [Kiro Crew](#)

VERCEL

Frontend cloud for deploying web apps and AI interfaces.

Step 1: Redeem your v0 credits

- Open v0.app and sign in, or create a free account.
- Click your profile picture (top right), then Credits, then Redeem Code.
- Enter your code to get \$30 in v0 credits. The code will be sent to all checked-in attendees at 10:30am.

Step 2: Build your harness

- Prompt v0 to scaffold your project, e.g. "Build a Next.js dashboard that monitors my agent's context, memory and tool calls in real time."
- v0 uses the AI SDK under the hood, and you can ask it to wire up agent loops, tool calling and streaming.
- Iterate with follow-up prompts until it works the way you want.

Step 3: Deploy and demo

- Click Deploy to get a live, publicly accessible URL instantly.
- Drop the link in your submission so judges can try it live.

More resources

- AI SDK (agent harness layer): ai-sdk.dev
- AI Gateway: vercel.com/docs/ai-gateway
- v0 docs: v0.app/docs

OPENAI

1250 codex credits for the hackathon.

- **Access:** Checked-in participants will be sent a code to their registered email address from 10:30am on Saturday.

VOYAGE AI

[Voyage AI](#) builds state-of-the-art embedding models and rerankers for accurate, efficient search over unstructured data — leading the field in retrieval quality, model size, vector dimensions, and latency. Every participant gets 200 million free tokens to use for the event, which should cover most projects without incurring any cost.

Get Set Up

- [Create a Voyage AI Account](#) — Sign up with just an email address.
- [Generate an API Key](#) — From Organization > API Keys, create a new secret key and save it immediately; it won't be shown again.
- [Add a Payment Method](#) — Unlocks standard Tier 1 rate limits. No charge is required, and you won't be billed as long as you stay within your free token allowance. See the [rate limits page](#) for details.

Build with the API

- [API Key & Python Client Setup](#) — Install the client with `pip install voyageai` and authenticate.
- [Quickstart Tutorial](#) — A step-by-step guide to building a simple RAG chatbot with Voyage embeddings, covering embedding text, vector search, and reranking.
- [Pricing & Free Credits](#) — Details on token costs and how the free allowance applies.

Track Usage & Get Support

- **Usage Dashboard** — Monitor tokens used and remaining free credits in real time.
- **Need more tokens?** 200M is generous, but if you're approaching the limit, ask the event host or any member of the Voyage AI team for a top-up — you won't run out during the event.
- **Have questions?** The Voyage AI team is on-site throughout the event. Don't hesitate to ask.